M. Thufail Alwannabil Samas

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SUMMARY

Mathematics graduate and BNSP-certified Data Scientist with experience in data analysis, machine learning, time-series analysis, and Python/PostgreSQL data pipelines. Developed monitoring, forecasting, and classification workflows, with a first-author peer-reviewed publication on XGBoost-based rainfall forecasting.

EXPERIENCE

PT Aplikanusa Lintasarta **Surabaya, Indonesia**
Junior Programmer January 2026 – April 2026

- Developed a project-stage DWDM optical monitoring and assurance analytics system for approximately 400 directional sensors, integrating 5-minute PRTG telemetry with ENIMS network metadata to identify operational failures, persistent signal changes, gradual degradation, and sensor priorities.
- Built an incremental Python and PostgreSQL time-series pipeline for ENIMS metadata synchronization, PRTG telemetry collection, gap repair, processed-observation generation, event detection, paired-sensor correlation, priority ranking, and API-served dashboard outputs.
- Implemented data-quality and failure-detection logic for missing or empty observations, low optical-power values, downtime, and zero coverage, collapsing adjacent 5-minute failure buckets into contiguous events while excluding invalid observations from statistical analysis.
- Developed sensor-adaptive change-point and gradual-degradation detectors using robust 6-hour medians, penalized PELT segmentation, Theil-Sen slopes, and serial-correlation-corrected Mann-Kendall tests, with regime and step-dominance checks to distinguish persistent signal changes from telemetry noise.
- Built a Python-backed HTML/CSS/JavaScript dashboard with Operations Board, Event Explorer, and Management Overview views for active failures, statistical events, paired-sensor evidence, investigation charts, and current sensor-priority ranking.

SMP GIKI 3 Surabaya **Surabaya, Indonesia**
Coding and AI Teacher August 2025 – December 2025

- Taught Python fundamentals to Grade 8 and 9 students over 5 months, covering variables, data types, operators, conditionals, loops, and programming logic through practical coding exercises.
- Introduced foundational AI concepts using Teachable Machine, guiding students through datasets, model training, and image classification through hands-on experimentation.
- Designed lesson materials, coding exercises, quizzes, and sample code from scratch, using practical programming activities and assessments to evaluate student understanding.

Stasiun Meteorologi Maritim Kelas II Tanjung Perak Surabaya (BMKG) **Surabaya, Indonesia**
Data Science Intern February 2024 – June 2024

- Developed a one-page extreme-weather early warning dashboard using HTML, CSS, JavaScript, PHP, MySQL, and API-based periodic updates for BMKG's internal forecasting workflow.
- Designed monitoring views for current weather conditions, warning indicators, tables, charts, and filters, consolidating key information into a single operational interface.
- Processed and compared BMKG cloud-prediction imagery with actual Himawari IR satellite imagery using Python, supporting descriptive validation of cloud movement and temperature patterns.
- Developed a 3-layer BiLSTM rainfall forecasting model using 7-day sequences and an 80:20 chronological split, achieving MAAPE of 0.8073 and RMSE of 10.2734 mm.

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA) **Surabaya, Indonesia**
Programming Teaching Assistant August 2022 – July 2024

- Guided undergraduate students across 4 semesters of Basic Programming and Programming courses in MATLAB, Python, and computational problem-solving.
- Provided hands-on guidance in debugging, data handling, and algorithmic thinking, helping students translate mathematical and computational concepts into working code.

- Developed introductory machine-learning materials covering 3 algorithms – Linear Regression, K-Nearest Neighbors, and K-Means clustering – and provided structured technical feedback on student assignments.

EDUCATION

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA) — **Surabaya, Indonesia**
Bachelor of Mathematics (S.Mat.) — August 2021 – January 2025

- **GPA:** 3.46/4.00
- **Undergraduate Thesis:** Rainfall Forecasting in Sumenep Regency Using XGBoost and Grid Search.
- **Academic Highlight:** Completed the 4-year bachelor's program in 7 semesters (3.5 years).
- **Relevant Coursework:** Artificial Intelligence, Programming, Mathematical Statistics, Statistical Methods, Multivariate Analysis, and Applied Linear Algebra.

SKILLS

- **Programming & Query:** Python, SQL, R, MATLAB, PHP
- **Data Engineering:** PostgreSQL, MySQL, Data Ingestion, Data Validation, Data Transformation, Data Pipelines
- **Libraries & Frameworks:** NumPy, Pandas, Scikit-learn, XGBoost, Statsmodels, TensorFlow/Keras, Matplotlib, Seaborn
- **Analytics & Modeling:** Data Analysis, Data Preprocessing, Feature Engineering, Statistical Modeling, Machine Learning, Deep Learning, Model Evaluation, Time Series Analysis & Forecasting
- **Development & Visualization:** Jupyter Notebook, Git, GitHub, REST API, HTML, CSS, JavaScript, Dashboard Development

PROJECTS

Rainfall Forecasting with XGBoost and Grid Search
Built an XGBoost next-day rainfall forecasting workflow using 1-day lagged BMKG weather features, fold-wise Min-Max scaling, 10-fold TimeSeriesSplit, and Grid Search, achieving MAAPE of 0.9152 and RMSE of 11.9566 mm.

Thyroid Cancer Recurrence Classification with XGBoost, Information Gain, and Grid Search
Built an XGBoost recurrence-classification workflow using the 383-record UCI dataset, fold-wise Information Gain, 10-fold cross-validation, and Grid Search, achieving 97.25% accuracy, 95.37% F1-score, 98.05% specificity, and 98.60% ROC-AUC.

Rainfall Forecasting with BiLSTM and Grid Search
Built a 3-layer BiLSTM next-day rainfall forecasting workflow using 7-day multivariate sequences, an 80:20 chronological split, Min-Max normalization, and Grid Search, achieving MAAPE of 0.8073 and RMSE of 10.2734 mm.

Rice Price Forecasting with ARIMA and Walk-Forward Validation
Built two ARIMA forecasting models for 60 months of East Java medium and premium rice prices using ADF-based differencing, residual diagnostics, and expanding-window walk-forward validation, achieving MAPE of 2.081% and 1.8174%, respectively.

PUBLICATIONS

Prediksi Curah Hujan di Kabupaten Sumenep Menggunakan Metode Extreme Gradient Boosting (XGBoost) dan Algoritma Grid Search
First Author | Peer-Reviewed Article — *MIND Journal*, Vol. 11, No. 1, pp. 30 – 43, June 2026
DOI: 10.26760/mindjournal.v11i1.30-43

CERTIFICATIONS

- **Data Scientist** — Badan Nasional Sertifikasi Profesi (BNSP) | September 2024 – September 2027